ME 137 Project

By: Ashley Donecker,
Jesus Serafin, and Hope Lazo

Goals of our project

1. Create a creative design
2. Create an efficient fruit removal tool
3. Make it mobile so it could move around the field easier

1

Initial Design Plans

Key Ideas

1

Mobile

We want our design to be able to move around the field freely. For this we put wheel and a tractor hitch

2

Movement

We know that trees come in all shapes and sizes to height was a factor in our design. To create this we used a scissor lift to lift the mechanism up.

3

Safe

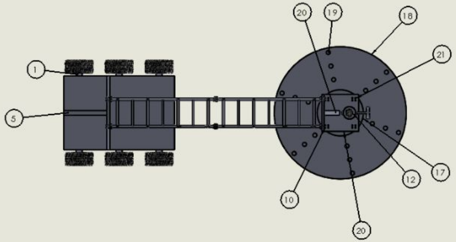
The mechanism is safe to the tree it oscillates at a fixed rate so it doesn’t damage the tree.

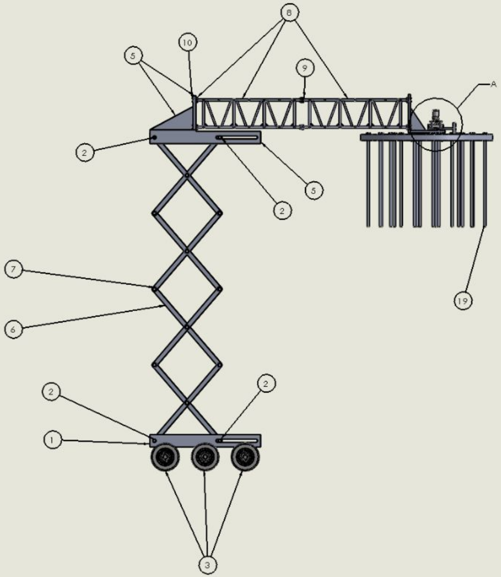
Design Explained

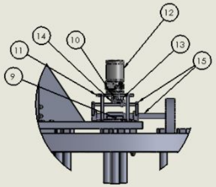
Reason for design, the Jellyfish with the rods as it oscillated helps nock the fruit from the tree off the tree. The whole contraption acts like a hair brush that goes into the fruit tree, but by oscillating motion instead of brushing helps remove the fruit as the rods hit the fruit. The scissor lift helps as a variable to any size tree so that it can work with any tree.

2

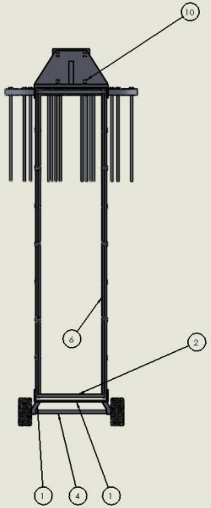
Full Assembly Drawing

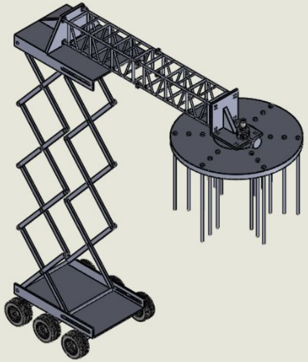






DETAIL A
SCALE 1 : 10



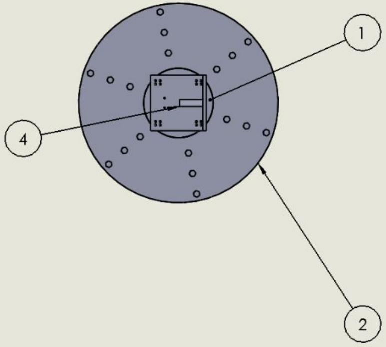


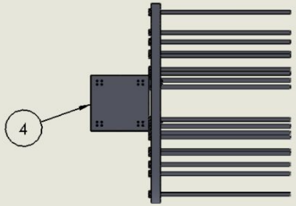
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Bottom SL		1
2	Beam SL		4
3	2181T12	Steel-Hub Pneumatic Trailer Wheel for Highway Use	6
4	Wheel Atatchment SL		3
5	Top SL		1
6	Link		16
7	Bolt SL		20
8	Truss 1		2
9	91257A722	Zinc Yellow-Chromate Plated Hex Head Screw	18
10	90767A114	High-Strength Steel Cap Nut	68
11	Housing Motor		1
12	6299K55	7.3 hp Hydraulic Motor	1
13	Rod converter		1
14	Motor Attachment		1
15	Double Rod		1
16	92620A716	Zinc Yellow-Chromate Plated Hex Head Screw	2
17	Middle JF		1
18	Main JF		1
19	Rod JF		18
20	Truss Connection JF		1
21	91257A738	Zinc Yellow-Chromate Plated Hex Head Screw	16
22	91257A726	Zinc Yellow-Chromate Plated Hex Head Screw	16
23	91257A724	Zinc Yellow-Chromate Plated Hex Head Screw	16

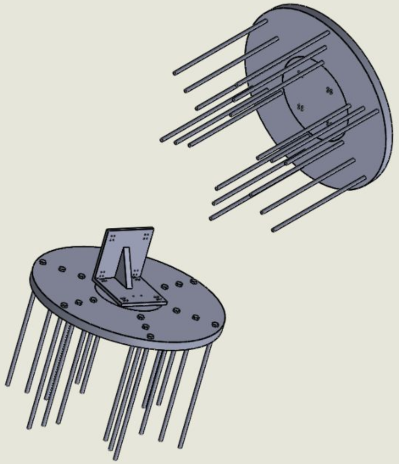
Full Assembly DRW

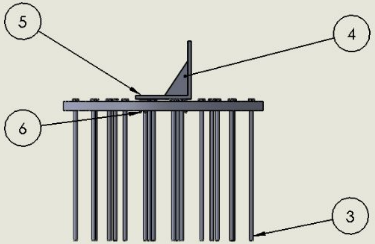
3

Jellyfish Assembly Drawing









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1	Middle JF		1
2	Main JF		1
3	Rod JF		18
4	Truss Connection JF		1
5	91257A738	Zinc Yellow-Chromate Plated Hex Head Screw	16
6	90767A114	High-Strength Steel Cap Nut	16

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

Jellyfish DRW

A3

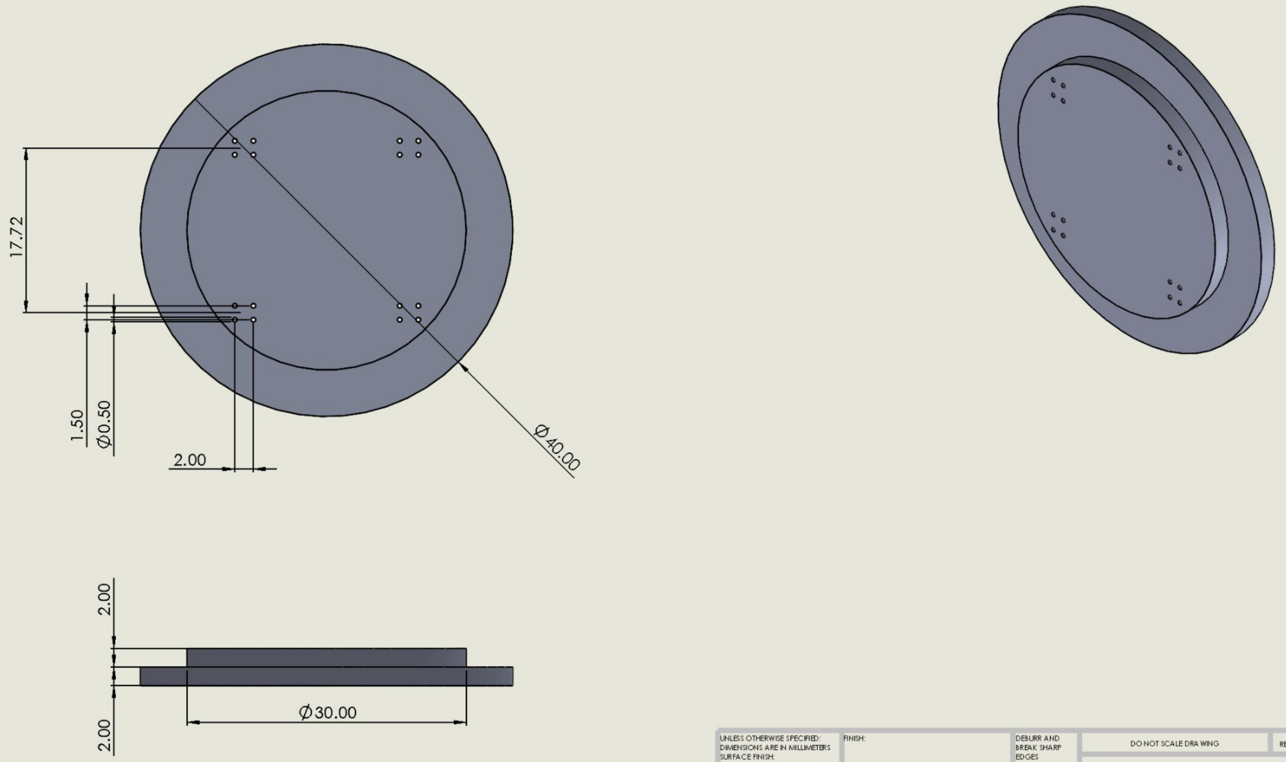
WEIGHT:

SCALE:1:40

SHEET 1 OF 7

4

Jellyfish Components Drawings



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ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

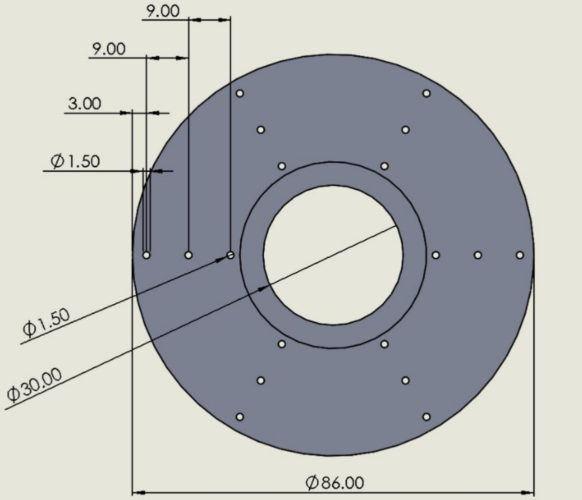
Middle JF

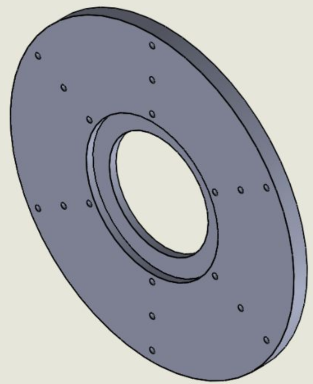
A3

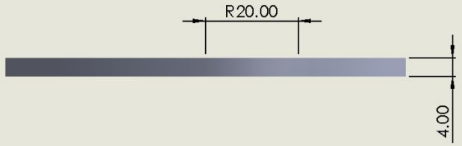
WEIGHT:

SCALE:1:10

SHEET 2 OF 7







Sheet3

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DRAWN					
CHK'D					
APPV'D					
MFG					
Q.A			MATERIAL:	DWG NO. Main JF	A3
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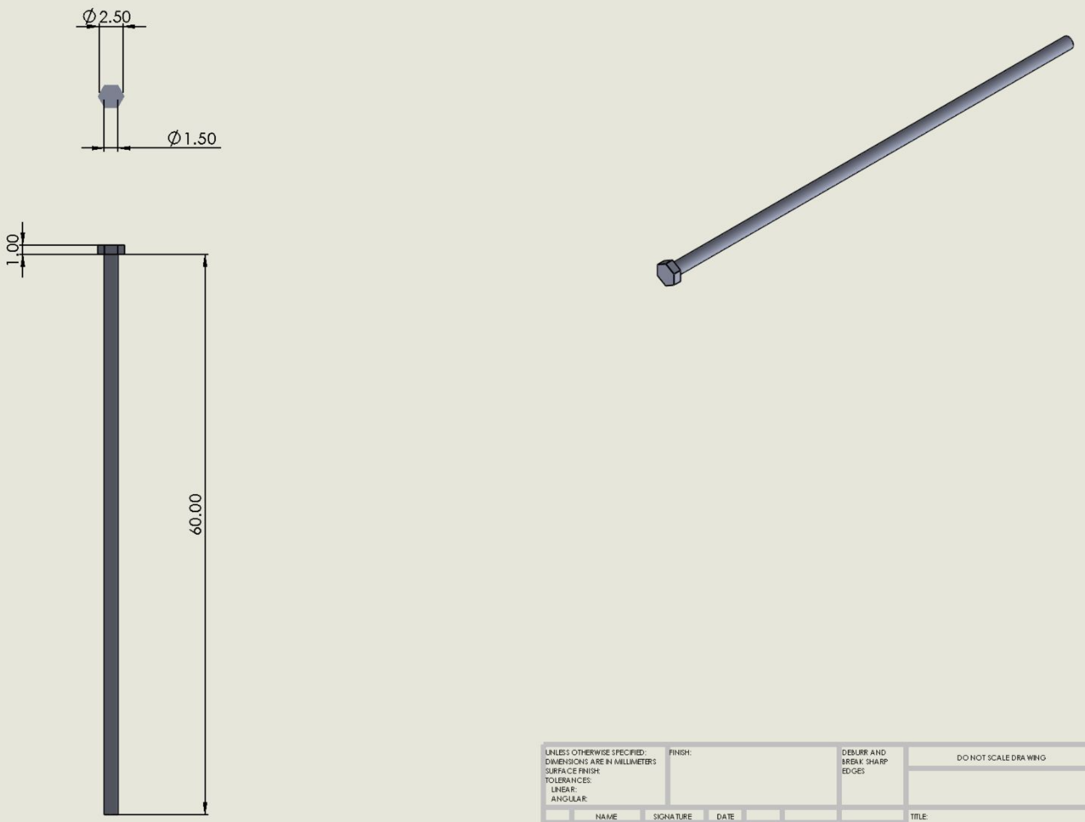

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FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE		
DRAWN					
CHK'D					
APPV'D					
MFG					
Q.A					

TITLE:

MATERIAL:

DWG NO.

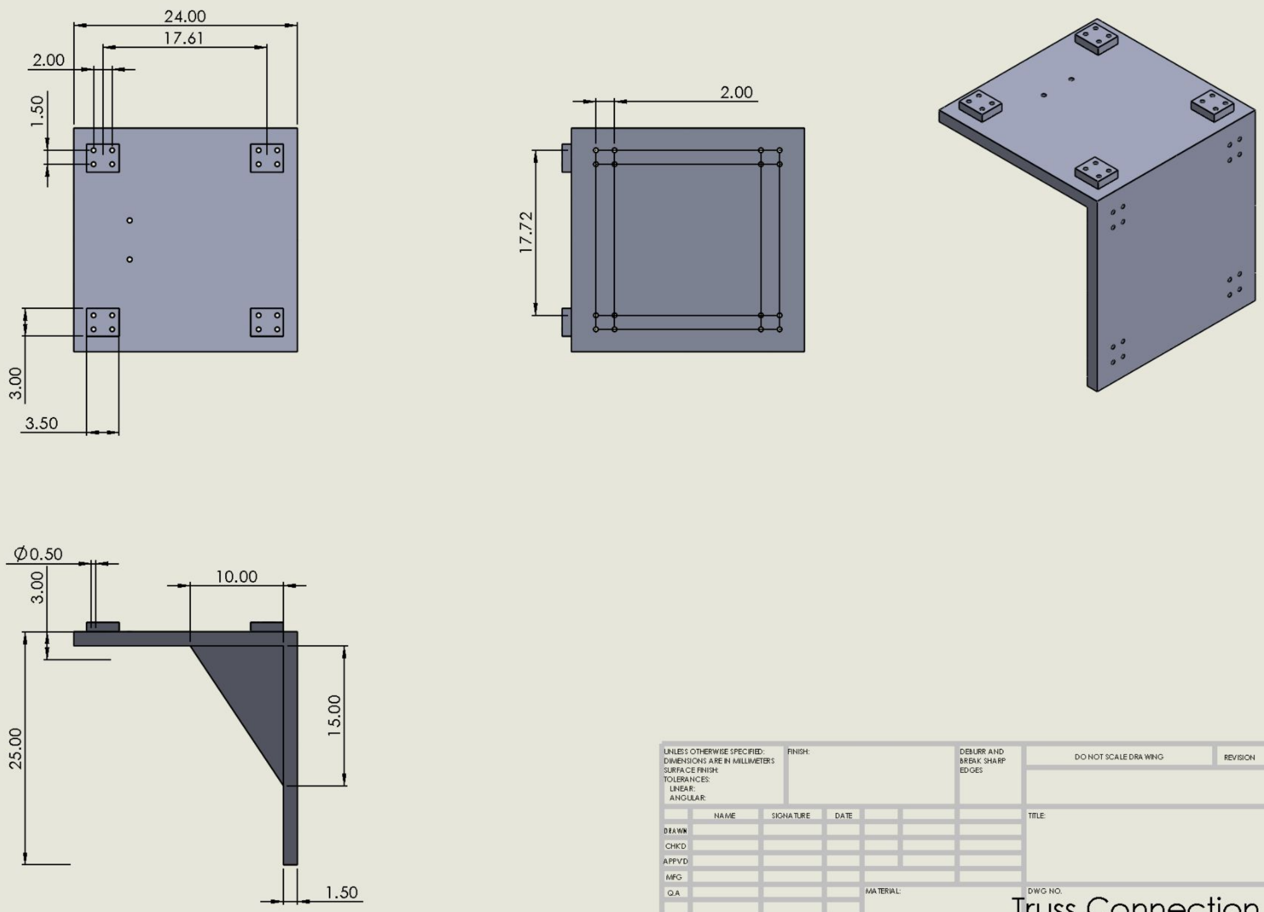
Rod JF

A3

WEIGHT:

SCALE:1:10

SHEET 4 OF 7



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FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

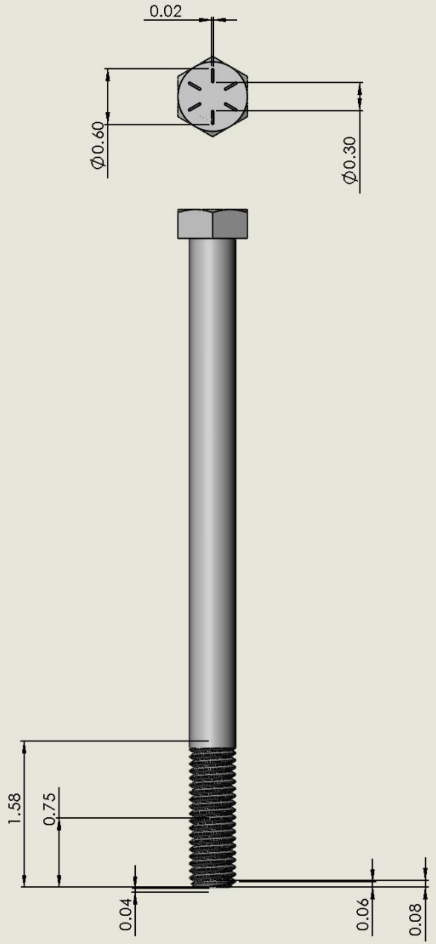
Truss Connection JF

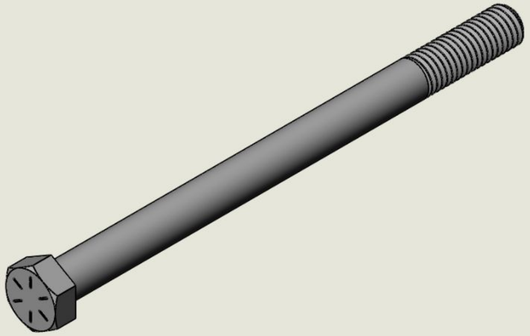
A3

WEIGHT:

SCALE:1:10

SHEET 5 OF 7





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FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

WEIGHT:

DWG NO.

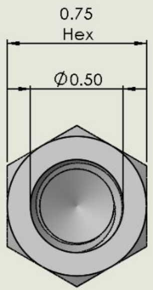
Hex Head Screw

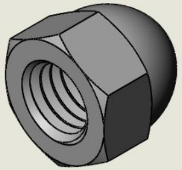
A3

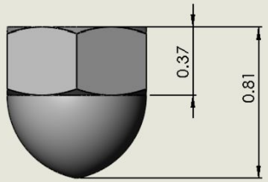
SCALE:1:1

SHEET 6 OF 7

Sheet7







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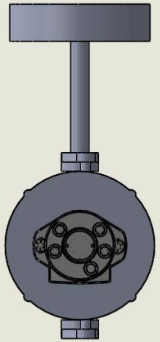
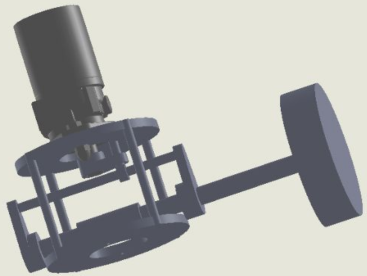
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO.	A3
				WEIGHT:		SCALE:2:1	SHEET 7 OF 7

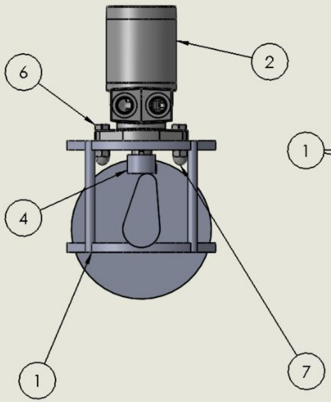
High-Strength Steel Cap Nut

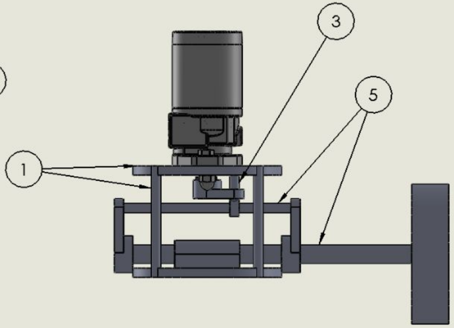
5

Oscillation Machine Assembly Drawing









ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Housing Motor		1
2	6299K55	7.3 hp Hydraulic Motor	1
3	Rod converter		1
4	Motor Attachment		1
5	Double Rod		1
6	92620A716	Zinc Yellow-Chromate Plated Hex Head Screw	2
7	90767A114	High-Strength Steel Cap Nut	2

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

Oscillation Machine DRW

A3

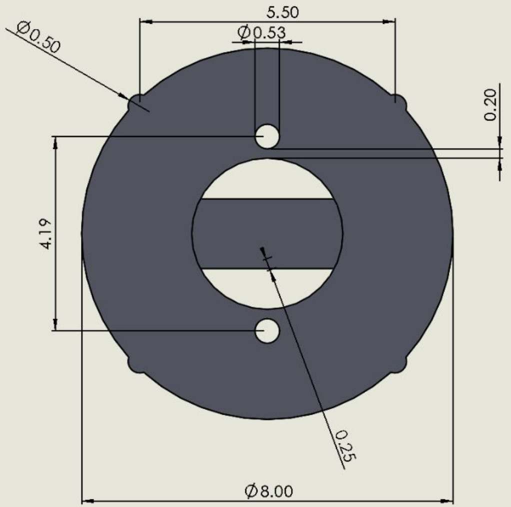
WEIGHT:

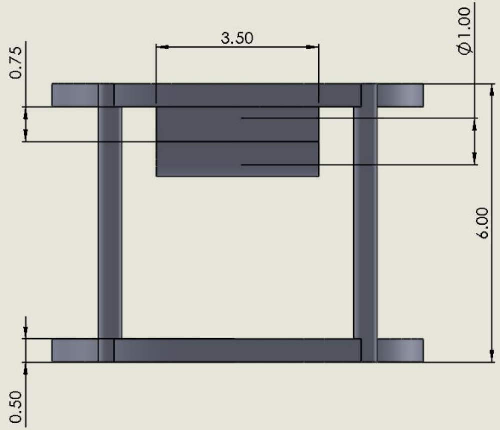
SCALE:1:5

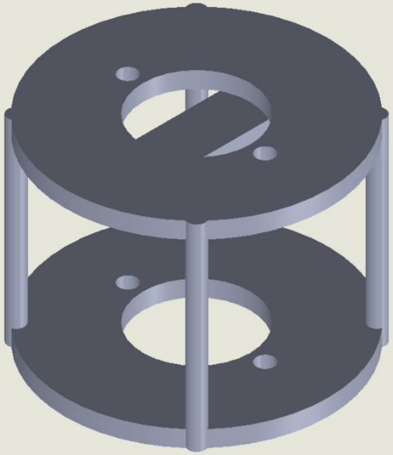
SHEET 1 OF 7

6

Oscillation Machine Components Drawings







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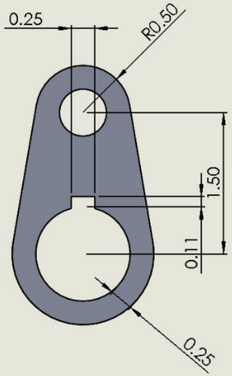
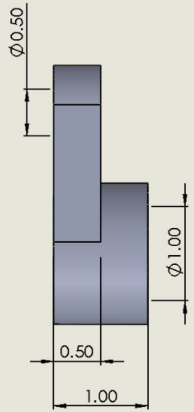
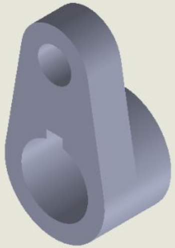
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Housing Motor	A3
				WEIGHT:		SCALE1:2	SHEET 2 OF 7







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	NAME	SIGNATURE	DATE	TITLE:	
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CHK'D					
APPV'D					
MFG					
Q.A			MATERIAL:	DWG NO. Motor Attachment	A3
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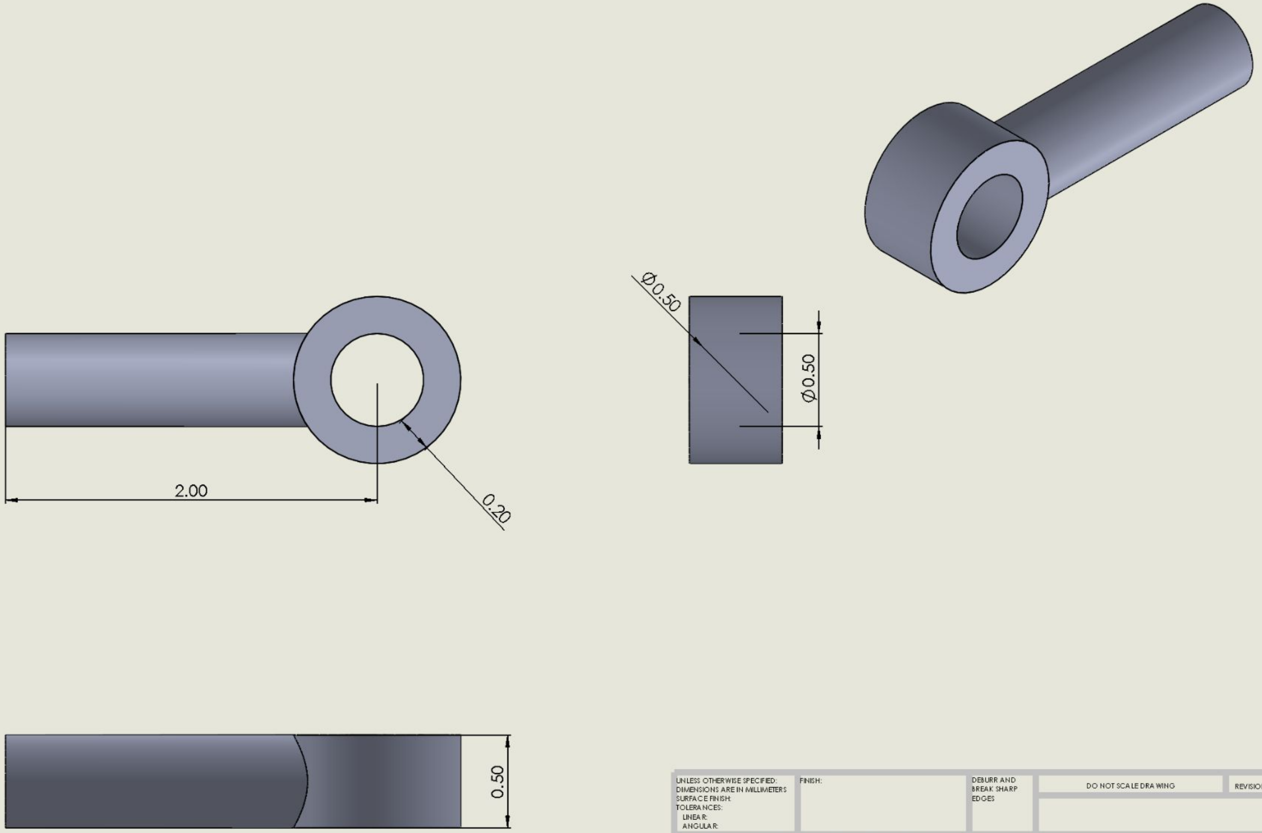

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LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

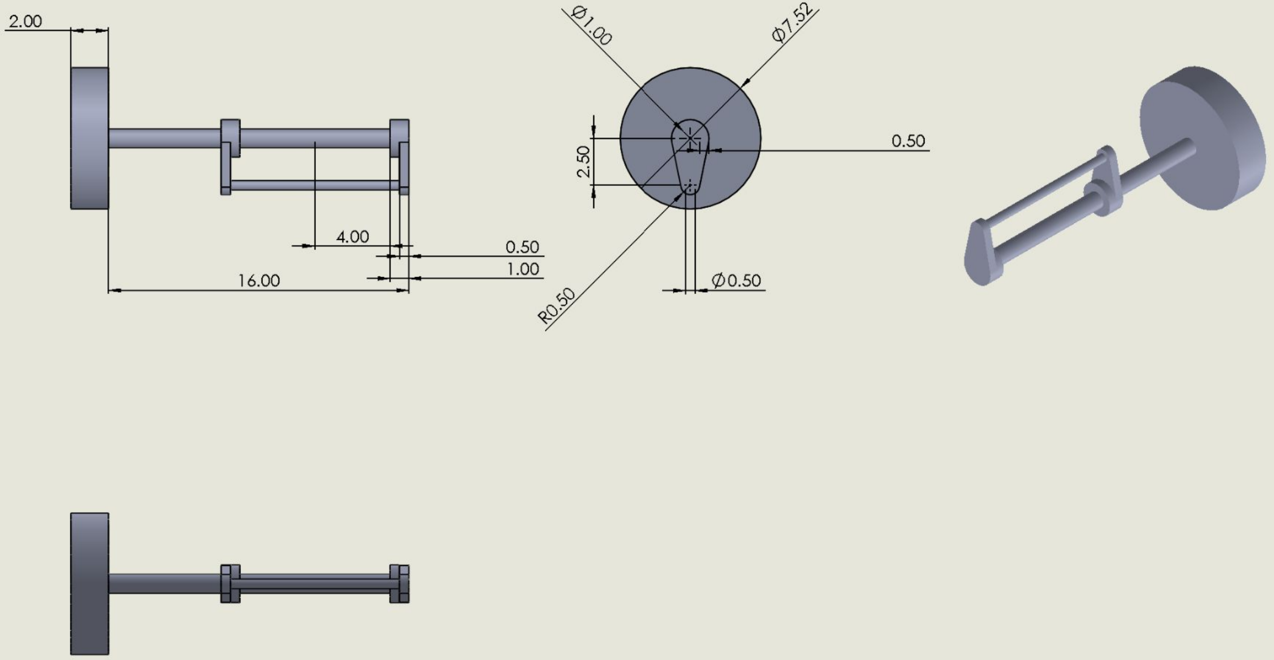
DWG NO.

Rod Converter

A3

SCALE:2:1

SHEET 4 OF 7



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TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

Double Rod

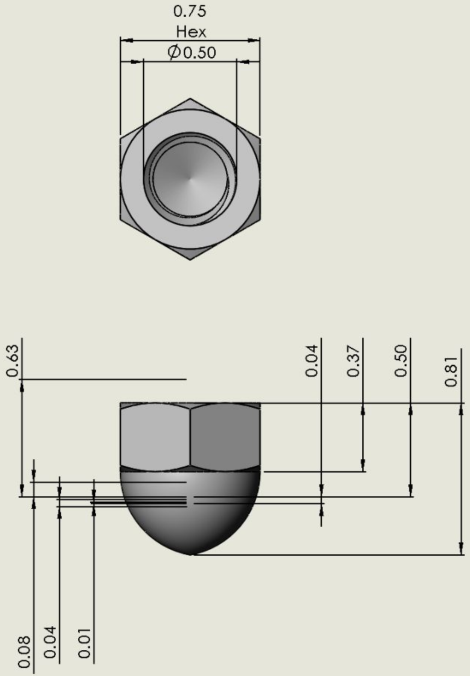
A3

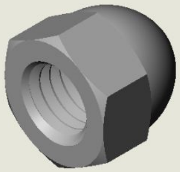
WEIGHT:

SCALE:1:5

SHEET 5 OF 7

Sheet6





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	NAME	SIGNATURE	DATE			TITLE:	
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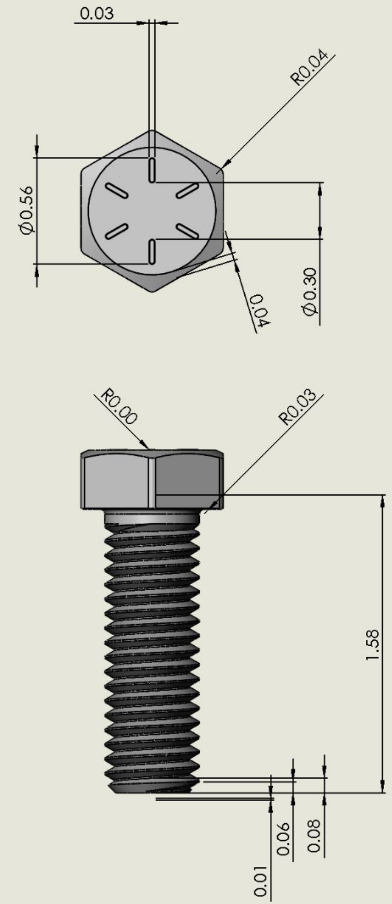
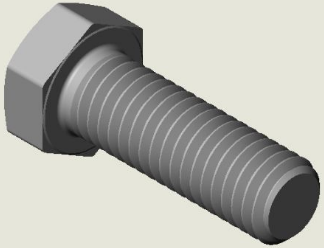
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO.	A3
				WEIGHT:		SCALE:2:1	SHEET 6 OF 7

High-Strength Steel Cap Nut





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SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

WEIGHT:

DWG NO.

Hex Head Screw

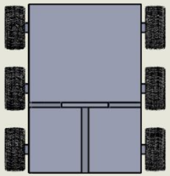
A3

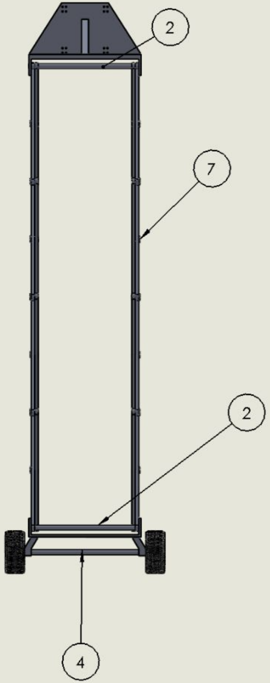
SCALE:2:1

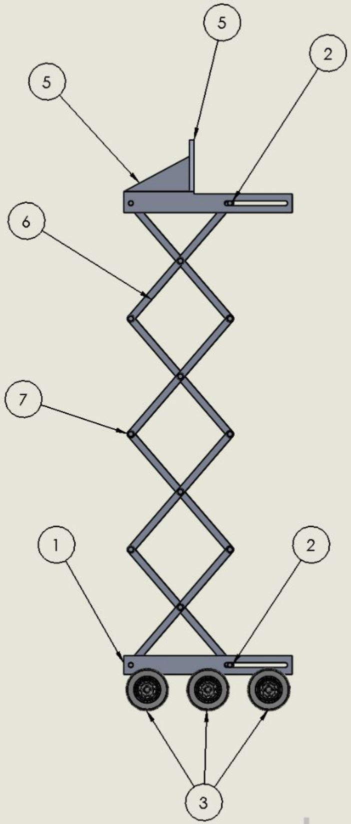
SHEET 7 OF 7

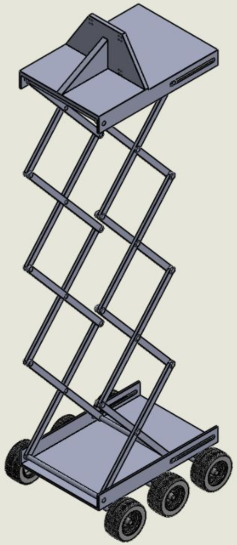
7

Scissor Lift Assembly Drawing









ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Bottom SL		1
2	Beam SL		4
3	2181T12	Steel-Hub Pneumatic Trailer Wheel for Highway Use	6
4	Wheel Atatchment SL		3
5	Top SL		1
6	Link		16
7	Bolt SL		20

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

MATERIAL:

WEIGHT:

TITLE:

DWG NO.

Scissor Lift DRW

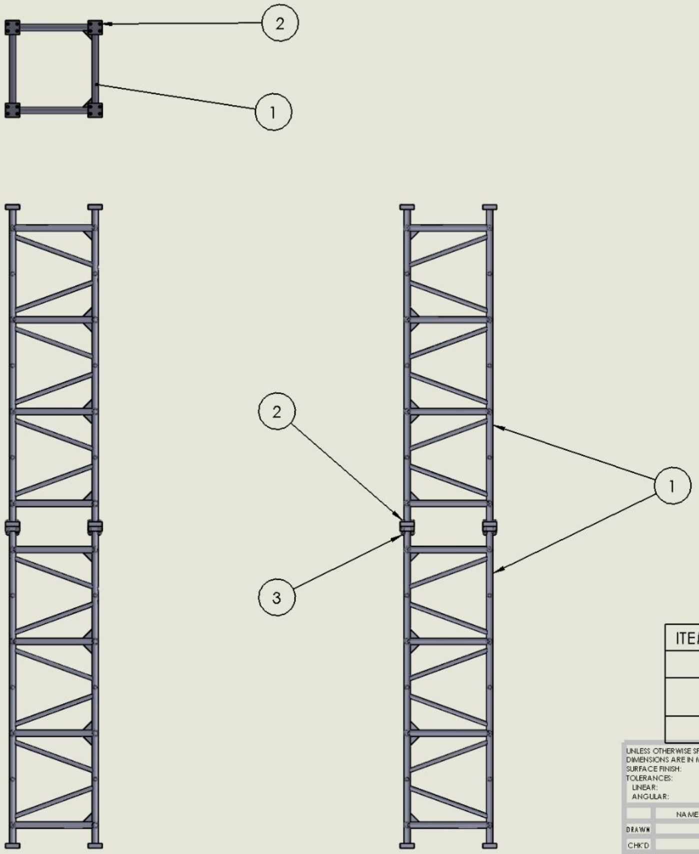
A3

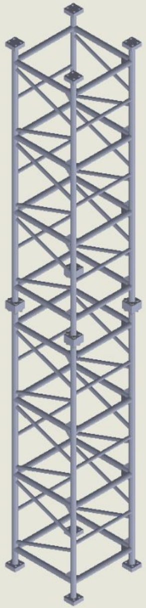
SCALE:1:40

SHEET 1 OF 1

8

Truss Assembly Drawing





ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Truss 1		2
2	91257A722	Zinc Yellow-Chromate Plated Hex Head Screw	16
3	90767A114	High-Strength Steel Cap Nut	16

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO.

Truss DRW

A3

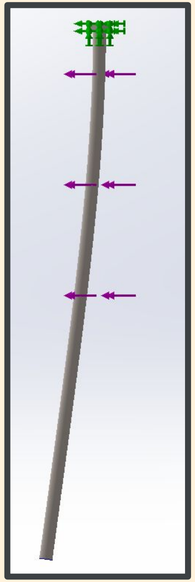
WEIGHT:

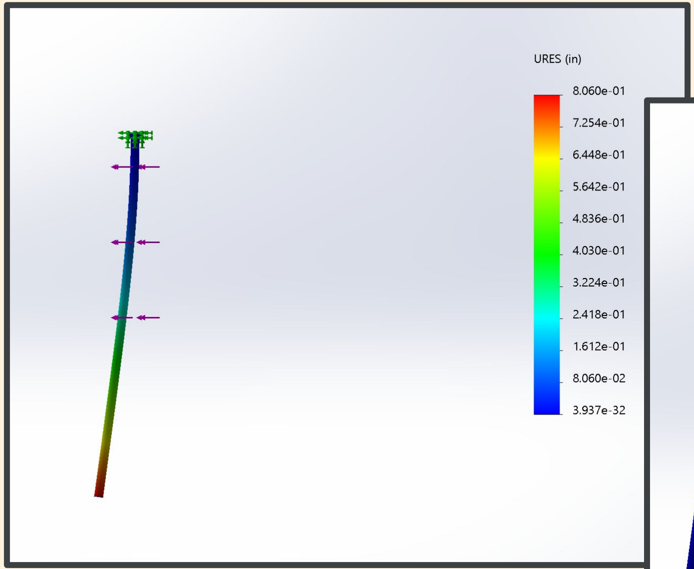
SCALE:1:20

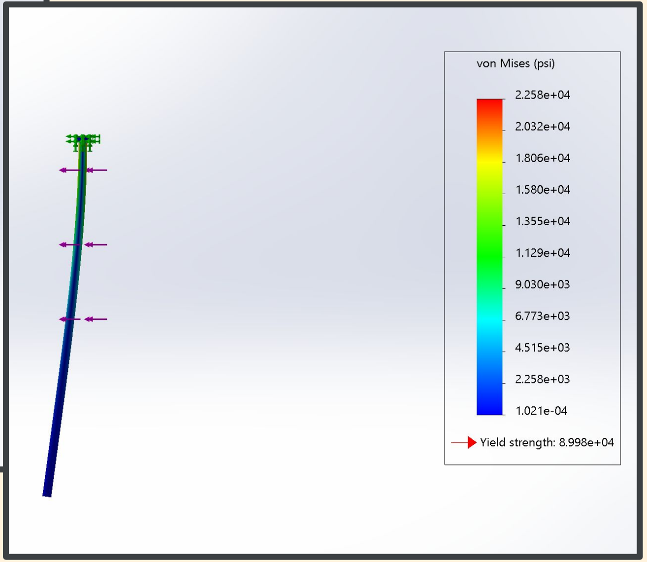
SHEET 1 OF 1

Stress Analysis Assumption

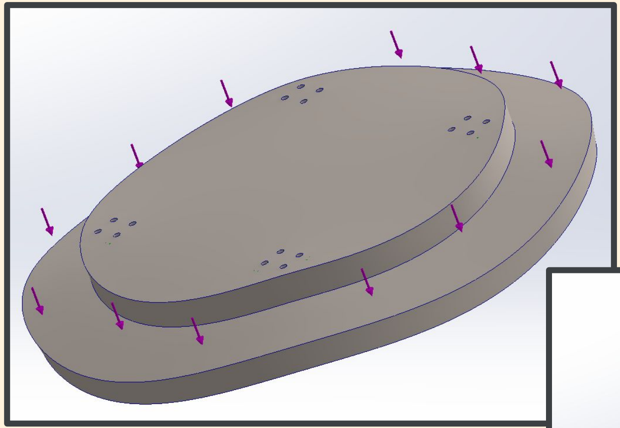
We assume the force on the rods are 1 kN from google

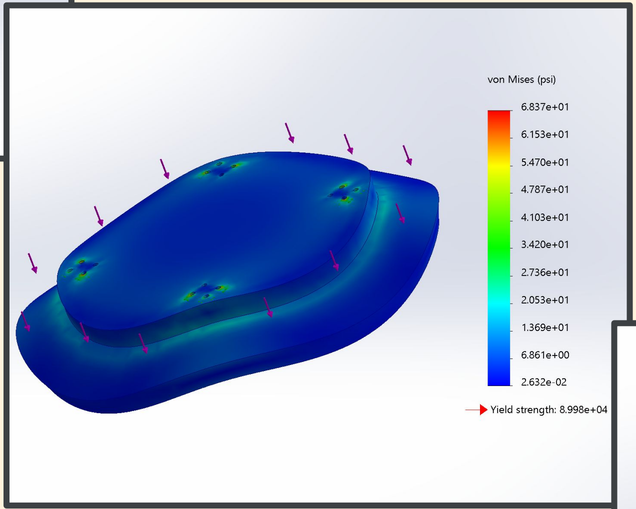


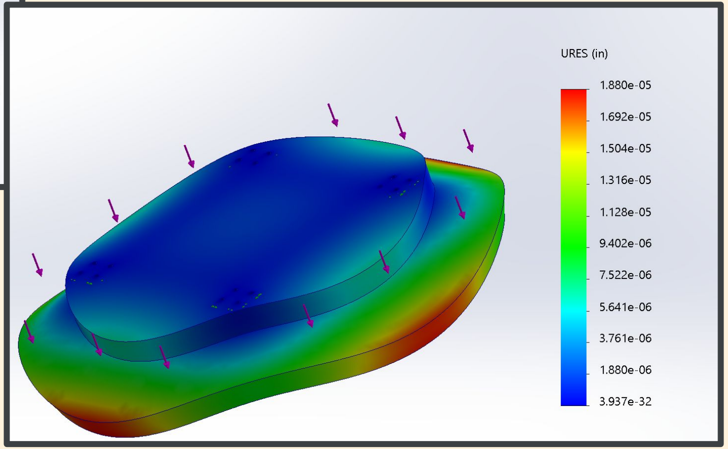




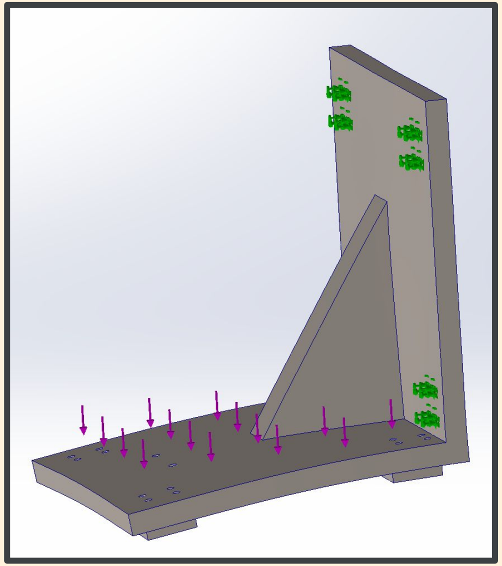
We assume the force on the Middle JF is 500LB based on Solidworks Mass Properties



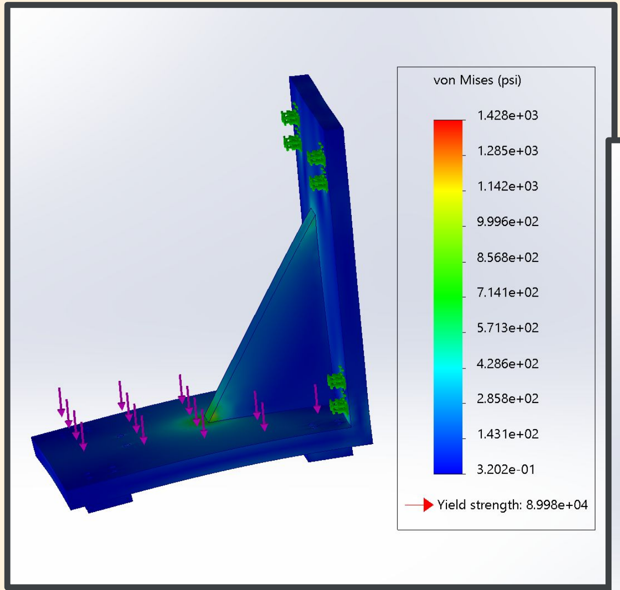


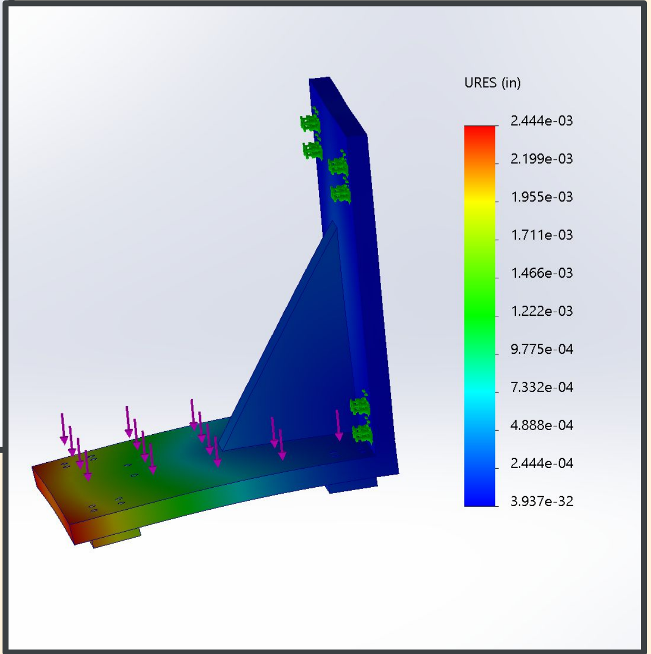


Stress Analysis Assumption



We assume the force on the Truss Connection JF





Stress Analysis Assumption

Full Assembly Animation Video

Thank you :)

IMPORTANT NOTE:
We are sorry but we might have misunderstood the project a little bit. That is why you might have seen that our main design looks a little different than the one posted in the announcements in the pdf. We are a group of three. And we got confirmation we could be a group of three if we did something a little extra, maybe we took that interpretation a little too far when being creative. We hope that you like our different design and see it being very creative. We spent a lot of time and hope you can appreciate what we did. And the reason why we have the scissor lift was because we confirmed with the Professor that we would have a connection device with the fruit removal contraption with the group/tractor. Thank you for your time to read this!!!